

Differences Between Containers and Virtual Machines

Objectives

After completing this reading, you will be able to:

- Define containers and virtual machines
- Explain the difference between containers and virtual machines

Introduction

When working with virtualization tools, it is important to understand the differences between containers and virtual machines or VMs to effectively choose the appropriate virtualization technology for your applications and workloads. Here's a brief guide outlining the key distinctions.

Feature	Containers	Virtual machines
Definition	A lightweight method of virtualization that allows for running multiple isolated user-space instances	A more traditional, hardware-based virtualization approach that creates a full virtual replica of a real server
System overhead	Uses minimal system resources, leading to higher efficiency and density	Requires more system resources due to the need for a full operating system for each VM
Startup time	Containers start almost instantaneously, providing fast scalability	VMs take longer to boot since they must start an entire operating system
Performance	Close to native performance due to sharing the host system's kernel	Slightly reduced performance compared to native and containers due to hardware emulation
Isolation	Provides process isolation, which is lighter but less isolated than VMs	Offers strong isolation by emulating physical hardware, resulting in separate kernels for each VM
Use case	Used for consolidating physical servers, hosting multiple applications on a single machine, and running different operating systems for testing purposes	Ideal for businesses looking for scalable solutions, such as web hosting, extensive data analysis, and cloud-based applications

Summary

In this reading, you learned that:

- Containers perform the lightweight virtualization method, run multiple isolated user-space instances, and require minimal system resources. It starts instantaneously. The containers' performance is close to native and provides process isolation, which is lighter but less isolated than VMs.
- Virtual machines use a traditional hardware-based virtualization approach to create a full virtual replica of a real server and require more system resources. It takes longer to boot. It uses hardware emulation, and so provides reduced performance.
- Containers are ideal for microservice use cases. However, the VMs are suited for applications requiring full isolation, full OS control, or running multiple applications on different OS.



Skills Network